Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Options and Futures

Song Xiao

xiaosong1@cuhk.edu.cn

Fall Term 2025

© S.Xiao (2025)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
▷ Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Volatility

© S.Xiao (2025)

Outline

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
▷ Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Volatility in Black-Scholes
- Evidence on volatility
- The implied volatility surface
- Local volatility models
- Stochastic volatility models

© S.Xiao (2025)

Volatility in Black-Scholes

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
▷ Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The volatility of the underlying is a key parameter of the BS model and a key input in the BS formula.
- By assumption, the underlying follows a GBM:

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

The (instantaneous) volatility σ is assumed to be constant.

- This implies that

$$\log(S_T/S_t) = (\mu - \sigma^2/2)(T-t) + \sigma(W_T - W_t).$$

The log return from t to T, $\log(S_T/S_t)$, is normally distributed, with variance $\sigma^2(T-t)$.

© S.Xiao (2025)

Estimating σ from historical data

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
▷ Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- If S follows a geometric Brownian motion with volatility σ, then $\log(S_{t+\Delta t}/S_t)$ has standard deviation $\sigma\sqrt{\Delta t}$.
- Suppose the reference time unit is 1 year. Then the annual log return, $\log(S_{t+1}/S_t)$, has standard deviation σ.
- Similarly, the daily log return,[1] $\log(S_{t+\frac{1}{250}}/S_t)$, has standard deviation $\frac{\sigma}{\sqrt{250}}$.
- Given an estimate of the standard deviation of the daily log return, we multiply by $\sqrt{250}$ to obtain the corresponding estimate of the standard deviation of the annual log return.

[1] By convention, one day corresponds to $1/250$ of a year (non-business days are not counted).

© S.Xiao (2025)

Estimating σ (cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
▷ Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Suppose we have daily historical data on the underlying price $S_{t_0}, S_{t_1}, \ldots, S_{t_N}$, with $t_{i+1} - t_i = \frac{1}{250}$ (1 day).
- We can compute the sample standard deviation of daily log returns:

$$\hat{\sigma}_d = \sqrt{\frac{\sum_{i=1}^{N} [\log(S_{t_i}/S_{t_{i-1}}) - \hat{m}_d]^2}{N-1}},$$

where

$$\hat{m}_d = \frac{\sum_{i=1}^{N} \log(S_{t_i}/S_{t_{i-1}})}{N}.$$

- The corresponding estimate $\hat{\sigma}$ of the annual volatility σ is given by $\hat{\sigma} = \sqrt{250}\hat{\sigma}_d$.

  –Example: if the sample standard deviation of daily log returns is 2%, then $\hat{\sigma} = 31.6\%$.

Volatility as a deterministic function of time

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
▷ Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Consider an extension of the BS model where volatility can vary over time in a deterministic manner, i.e.

$$dS_t = \mu S_t dt + \sigma_t S_t dW_t.$$

- Using Itô's lemma, we see that

$$d \log S_t = (\mu - \sigma_t^2/2)dt + \sigma_t dW_t.$$

- Integrating, we obtain

$$\log(S_T/S_t) = \int_t^T (\mu - \sigma_s^2/2)ds + \int_t^T \sigma_s dW_s.$$

- Since σ_s is non-stochastic, $\log(S_T/S_t)$ is normally distributed, with

$$\text{Var}[\log(S_T/S_t)] = \int_t^T \sigma_s^2 ds,$$

compared to $\text{Var}[\log(S_T/S_t)] = \sigma^2(T-t)$ in BS.

© S.Xiao (2025)

Extension (cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
▷ Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- **Result** [Merton (1973)]:
  The no-arbitrage price of a European option at time $t \leq T$ is obtained by plugging in the BS pricing formula a constant volatility parameter σ given by
  $$\sigma^2 := \frac{1}{T-t}\int_t^T \sigma_s^2 ds.$$
- This formula for σ^2 gives us the average instantaneous variance until expiration.

Link between stochastic volatility and non-normality

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
▷ Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- When volatility is not a deterministic function of time, the distribution of log returns over a discrete time interval is no longer normal.
- Suppose
$$dS_t = \mu S_t dt + \sigma_t S_t dW_t,$$
where σ_t is a stochastic process (as opposed to just a deterministic function of time).
- As in the case of deterministic σ_t, we have
$$\log(S_T/S_t) = \int_t^T (\mu - \sigma_s^2/2)ds + \int_t^T \sigma_s dW_s.$$
- But since σ_t is stochastic, $\log(S_T/S_t)$ is not normal.

Evidence on volatility

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
▷ Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

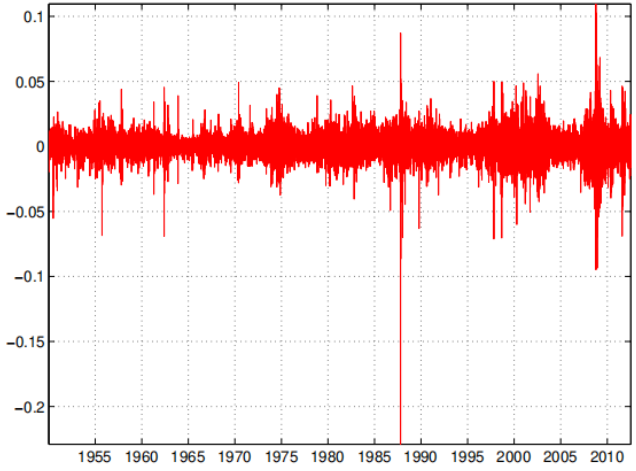
No matter whether you look at individual stocks, indices, commodities or currencies, the constant volatility assumption in BS is counterfactual.

- Volatility is time-varying and stochastic.
- Volatility exhibits mean reversion.
- Volatility typically increases when price falls: negative correlation between volatility and spot, referred to as asymmetric volatility.
- Changes in volatility are not just driven by movements of the underlying; there are specific shocks to volatility.
- The distribution of log returns is not normal
  - It exhibits fat tails and high/thin peaks.
  - For stocks and stock indices, it is negatively skewed.

S&P 500 returns

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
▷ Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Figure: SP500 Daily Returns: 1950 – 2012



S&P summary statistics

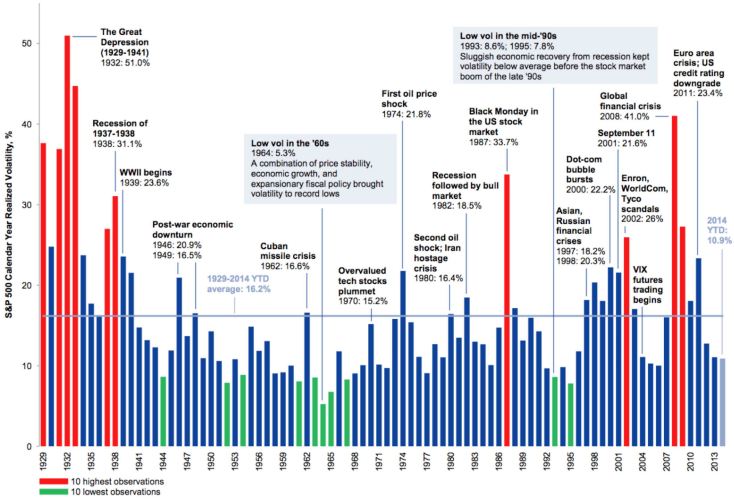
Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
▷ Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Daily log returns 03/01/1950–10/07/2012

Mean	0.028% (7.1% annualized)
StdDev	0.98% (15.4% annualized)
Min	-22.90%
Max	10.96%
Skewness	-1.03
Kurtosis	30.72
Autocorr(1)	2.90%
Autocorr(1) squared returns	14.17%

S&P 500 realized volatility

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
▷ Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B



A road-map

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
▷ Estimating σ
Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Since volatility is stochastic, the Black-Scholes formula does not provide accurate prices for standard options
  – However, it can be used as a lens for visualizing the cross section of observed prices of standard options via the notion of Black-Scholes implied volatility.
- For pricing exotic options, we can sidestep the difficulty of modeling volatility by deriving the risk-neutral distribution of the underlying from observed prices of standard options.
- A alternative approach is to model volatility:
  – Local volatility models;
  –Stochastic volatility models.

Implied volatility

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Implied volatility is a convenient way to express an option price.
- For instance, consider a European call struck at K with exercise date T which trades at price C^{mkt} at time t.
- Taking (S_t, t) as given, let $C^{\text{BS}}(\sigma)$ denote the BS price of the call as a function of σ, a shortcut for $C^{\text{BS}}(\sigma; S_t, t, r, T, K)$.
- The implied vol of the call is the level of volatility σ^{impl} such that

$$C^{\text{BS}}(\sigma^{\text{impl}}) = C^{\text{mkt}}.$$

- In other words, σ^{impl} is obtained by "inverting" the BS formula, which is possible because the function $C^{\text{BS}}(\cdot)$ is strictly increasing in σ.
  – There is no closed-form analytical expression for the inverse mapping.

The implied volatility surface

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- For a given underlying at a given point of time t, the volatility surface represents in 3D the implied volatility of options as a function of moneyness (captured by K or K/S) and time-to-expiration.
- In practice, since options are not traded for a continuum of strike prices and time-to-expiration, one needs to interpolate between the available points to obtain a smooth surface.
- Alternatively, implied volatilities can be represented in 2D, holding one argument fixed:

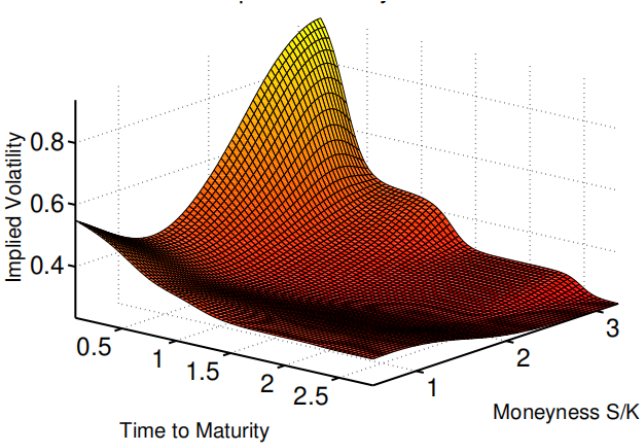
  – Fix time-to-maturity T and look at implied volatility as a function of the strike price $\rightarrow$ volatility curve.

  – Fix moneyness K or K/S and look at implied volatility as a function of time-to-maturity $T \rightarrow$ term structure of implied volatilities.

Implied volatility surface of S&P 500

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Figure: Implied Volatility Surface



Implied volatility from calls and puts

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- For a given underlying and for given (T, K), implied volatility can be equivalently obtained from European call or put options.
- Indeed, since both market and BS prices are arbitrage-free, they must satisfy put-call parity:

$$C^{\text{mkt}} = P^{\text{mkt}} + S - \text{PV}(K)$$

and

$$C^{\text{BS}}(\sigma) = P^{\text{BS}}(\sigma) + S - \text{PV}(K), \quad \text{for any } \sigma.$$

- Subtracting the first equation from the second:

$$C^{\text{BS}}(\sigma) - C^{\text{mkt}} = P^{\text{BS}}(\sigma) - P^{\text{mkt}}, \quad \text{for any } \sigma.$$

- Therefore, if σ is such that $C^{\text{BS}}(\sigma) = C^{\text{mkt}}$, it also satisfies $P^{\text{BS}}(\sigma) = P^{\text{mkt}}$ (and vice-versa).

Skew, smirk, smile

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

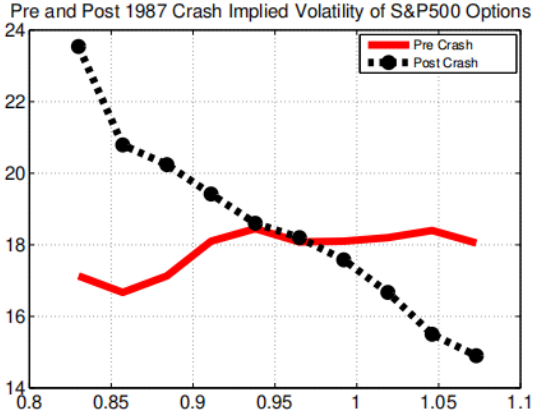
- Empirically, the shape of the volatility surface depends on the type of underlying (e.g. index vs. currencies) and it varies over time.
- Typical shape of the volatility curve for index options: Implied volatilities are decreasing in the strike price K. In other words, OTM puts have higher implied volatilities than OTM calls.
  - Consistent with a fat left tail of the distribution of the underlying.
  - This is referred to as a smirk  or a skew. Smile, by contrast, refers to a more symmetric U-shaped pattern[2]. 
- Typical finding in the time-to-expiration dimension: Implied vol is increasing in T when volatility has been low in recent period and decreasing in T when volatility has been high (which is consistent with mean-reverting volatility).

[2]The word "smile" can also refer loosely to any non-flat implied volatility curve.

© Steven Li (2024)

Implied volatility surface of S&P 500

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B



Source: Mark Rubinstein, Implied Binomial Trees, JF 1994

Before the 1987 crash, implied volatility curves tended to be more flat.

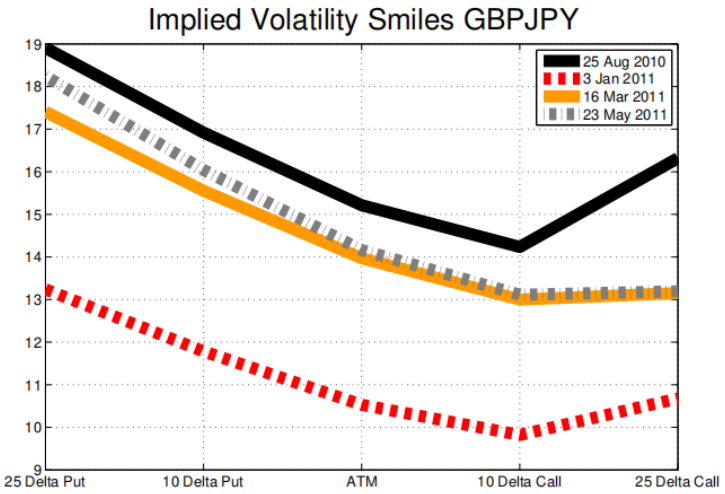
Volatility smile for currency options

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The underlying of a currency option is an exchange rate, i.e. the relative price of one currency relative to another.
- In principle, the underlying risks are symmetric – one currency can crash or the other.
- We often see high implied volatilities at both ends of moneyness (i.e. a smile pattern).

Volatility smile for the GBP/JPY exchange rate

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B



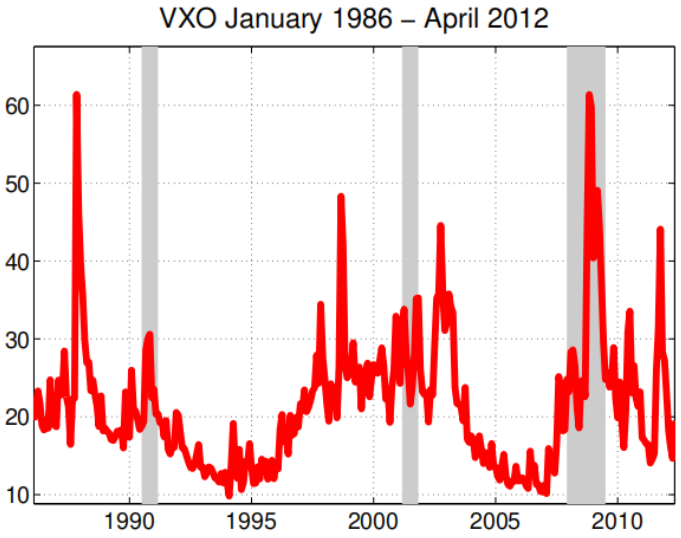
Dynamics of the volatility surface

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The implied volatility surface moves over time.
- Example for one point on the surface: The "old" VIX is the implied volatility of one-month ATM options on the S&P100 index.
  - The Figure on the next slide shows its evolution over period 1986–2012.
  - Largely driven by recently realized volatility.
- However, the volatility surface cannot have any arbitrary shape; it is constrained by no-arbitrage conditions (such as convexity of price wrt strike).

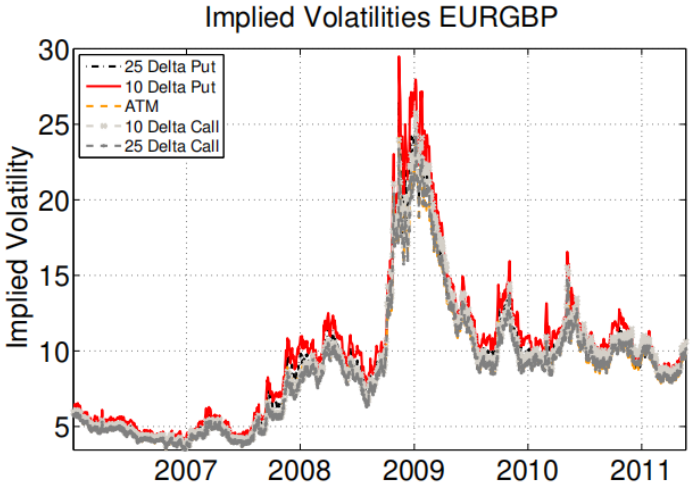
Old VIX (VXO) time series

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B



EUR/GBP implied volatility time series

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B



What implied volatilities say and do not say

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
▷ Implied volatility
Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The volatility surface provides a convenient way of visualizing the cross-section of option prices on a given underlying.
- The idea is to use the BS formula to express market prices in a convenient metric.
  – An implied volatility is nothing but "the wrong number to put in the wrong formula to obtain the right price" (Riccardo Rebonato).
- It is tempting to interpret an implied volatility as the market's best assessment of future volatility.
  – As it turns out, implied vol is not a good predictor of future realized vol, especially for longer horizons.

Implied risk-neutral distribution

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
▷ Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The cross-section of plain vanilla option prices (summarized by the implied volatility surface) says something about the risk-neutral distribution of the underlying.
- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q. The date 0 price of a call option as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x-K)\varphi(x)dx.$$

- **Result** [Breeden-Litzenberger (1978)] (proof in Appendix B):

$$\varphi(K) = e^{rT}C''(K).$$

- Note: This result is model-independent. It only uses the FTAP.

Breeden-Litzenberger formula: An example

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
▷ Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Let $S_{\frac{1}{12}}$ be the value of the S&P 500 index one month from now. This is a random variable from today's perspective, and we wish to derive its implied risk-neutral density φ.
- To do this we need today's market prices C for call options on the S&P 500 maturing in one month for a range of strike prices, $K_1 < K_2 < \ldots < K_n$.
- We estimate the function $C(K)$ for $K \in (K_1, K_n)$. This involves fitting a smooth curve through the observed data points $C(K_1), C(K_2), \ldots, C(K_n)$.
- We then obtain numerical estimates of $C''(K)$ for $K \in (K_1, K_n)$.
- The Breeden-Litzenberger formula gives us the value of φ for different values of $S_{\frac{1}{12}} \in (K_1, K_n)$. For example, for $S_{\frac{1}{12}} = 2000$, the value of φ is obtained from the second derivative of $C(K)$ at $K = 2000$:

$$\varphi(2000) = e^{\frac{r}{12}} C''(2000).$$

© S.Xiao (2025)

Implied risk-neutral distribution (cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
▷ Implied risk-neutral distribution
Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Once we have the implied risk-neutral density φ of S_T, we can calculate the date 0 price V of a derivative whose payoff is $g(S_T)$ at time T:

$$V = e^{-rT}\mathbb{E}^Q[g(S_T)] = e^{-rT}\int_0^\infty g(x)\varphi(x)dx.$$

- φ does not look lognormal
  - e.g. for index options, φ has heavier left tail than the lognormal distribution.
- Volatility smirk and negative skewness of the implied risk-neutral distribution are two sides of the same coin
  - Negative skewness is consistent with OTM puts being underpriced by BS relative to OTM calls.

Local volatility models

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- In a local volatility model, the price of the underlying is specified as follows:
$$dS_t = \mu S_t dt + \sigma(S_t, t) S_t dW_t.$$
- Under the risk-neutral probability Q,
$$dS_t = rS_t dt + \sigma(S_t, t) S_t dW_t^Q.$$
- In this model, volatility is stochastic, but the only source of randomness in volatility comes from the underlying. This is quite restrictive and certainly not realistic, though it can easily capture asymmetric volatility via an appropriate dependence on S.
- We can still form an instantaneously riskfree portfolio and derive the no-arbitrage PDE as in BS (though its solution is more complex).

Special case: Constant elasticity of volatility (CEV)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Specification of the price process of the underlying:

$$dS_t = \mu S_t dt + \sigma S_t^{\alpha} dW_t, \quad \alpha > 0.$$

  Then $\frac{dS_t}{S_t} = \mu dt + \sigma S_t^{\alpha-1} dW_t \rightarrow$ volatility at time t is $\sigma S_t^{\alpha-1}$, i.e. there is no direct dependence on t and the dependence on S is just a power function.
- When $\alpha = 1$, we are back to Black-Scholes (geometric Brownian motion).
- Relevant calibration: For $\alpha < 1$, volatility increases when price drops. This captures asymmetric volatility and has the potential to generate a skewed return distribution with a fat left tail.

CEV model (cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The dynamics of S under the risk-neutral probability Q are given by

$$dS_t = rS_t dt + \sigma S_t^{\alpha} dW_t^Q.$$

  The distribution of S_T under Q is known, though no longer lognormal.

- European call option prices can be obtained in closed-form for any α by computing $e^{-r(T-t)}\mathbb{E}_t^Q[(S_T - K)^+]$.
  – The pricing formulas can be expressed in terms of the noncentral chi-squared cumulative distribution function.

- For $\alpha < 1$, the model generates option prices that would produce a skew in the implied volatility curve.

- The model can be calibrated to have the best fit on option prices.
  – But there are only two parameters to choose, σ and α; not enough degrees of freedom to match all observed prices.

Local volatility function

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- In the general local volatility model, we have a function of two variables $\sigma(\cdot, \cdot)$ to play with – lots of degrees of freedom!
- Say you observe plain vanilla European call prices on a given underlying for different strike prices and times-to-expiration.
- Inverse problem: Under the assumption that the underlying price follows the process

$$dS_t = rS_t dt + \sigma(S_t, t) S_t dW_t^Q$$

under the risk-neutral measure Q, is there a function $\sigma(S, t)$ such that for all strikes and maturities, the model-predicted price matches the observed price, i.e. for all (K, T)

$$e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = C^{\text{mkt}}(K, T)?$$

The answer is: Yes!

Dupire's formula

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Market prices of plain vanilla European options are mapped into a local volatility function (also known as an implied volatility function or IVF).
- Given today's European call prices C, and the current price of the underlying S, the local volatility that will prevail at time T when $S_T = K$ is given by Dupire's formula:[3]

$$\sigma(K, T; S) = \frac{1}{S}\sqrt{\frac{\frac{\partial C}{\partial T}(K, T) + rS\frac{\partial C}{\partial K}(K, T)}{\frac{\partial^2 C}{\partial K^2}(K, T)}}$$

- Like the Breeden-Litzenberger formula, Dupire's formula is model-independent, relying only on the FTAP.
- By construction, the local volatility model using the IVF given by Dupire's formula matches plain vanilla option prices perfectly (and hence matches the observed volatility surface).

[3] This is **not** the BS implied volatility of an option with strike K and maturity T!

© Xiao (2024)

Dupire's formula: An example

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Suppose we wish to price one-month options on the S&P 500 index.
- To do this we need today's market prices C for call options on the S&P 500 for a range of strike prices, $K_1 < K_2 < \ldots < K_n$, and maturing at a range of dates, $T_1 < T_2 < \ldots < T_m \leq 1/12$.
- We estimate the function $C(K, T)$, for $K \in (K_1, K_n)$ and $T \in (T_1, T_m)$. This involves fitting a smooth surface through the observed data points $C(K_i, T_j), i = 1, \ldots, n; j = 1, \ldots, m$.
- We then obtain numerical estimates of $\frac{\partial C}{\partial T}$, $\frac{\partial C}{\partial K}$ and $\frac{\partial^2 C}{\partial K^2}$, for $K \in (K_1, K_n)$ and $T \in (T_1, T_m)$.
- Dupire's formula tells us what the value of σ is for different values of the index over the next one month. For example, if today's index value is 100, then the value of σ in one week's time at an index value of 115 at that time, is given by

$$\sigma(115, 1/52; 100) = \frac{1}{100}\sqrt{\frac{2\frac{\partial C}{\partial T}(115, 1/52) + 100r\frac{\partial C}{\partial K}(115, 1/52)}{\frac{\partial^2 C}{\partial K^2}(115, 1/52)}}$$

© S.Xiao (2025)

Local volatility models in practice

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The IVF can be used in conjunction with Monte-Carlo simulation for pricing exotic options:
  – Calibrate the local volatility function using current prices of plain vanilla options (thus matching the current volatility surface);
  –Simulate sample paths for $dS_t = rS_t dt + \sigma(St, t)StdW_t^Q$;
  –Compute the average option payoff and discount at the riskfree rate.
- This is the standard in the industry. Practitioners "price with a smile"
- Typically, the model is "recalibrated" every day on the new volatility surface (i.e. using the new set of plain vanilla prices).

Stochastic volatility models

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- A major limitation of local volatility models is that, even though volatility is stochastic, its evolution is entirely governed by the underlying. This is counterfactual.
- A more general stochastic volatility model is the following:

$$dS_t = \mu_t S_t dt + \sqrt{V_t} S_t dW_{St},$$

$$dV_t = a_t dt + b_t dW_{Vt},$$

with

$$dW_{St} dW_{Vt} = \rho dt.$$

- The second equation describes the stochastic evolution of the instantaneous variance V of the underlying, which is assumed to live a life of its own.
- Notice that there are two Brownian shocks, dW_S and dW_V, withinstantaneous correlation ρ.
- One commonly used approach is to specify the dynamics of S and V under Q and choose the parameters of these processes to provide the best possible fit for the observed volatility surface.

The Heston model

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- A very popular stochastic volatility model is that of Heston (1993).
- Under Q, the model is specified as follows:

$$dS_t = rS_t dt + \sqrt{V_t} S_t dW_{St},$$

$$dV_t = a(\bar{V} - V_t)dt + b\sqrt{V_t} dW_{Vt},$$

where the parameters a, b, and $\bar{V}$ are positive, and

$$dW_{St} dW_{Vt} = \rho dt.$$

- The instantaneous variance V follows a mean-reverting square-root process.
  - This process always remains positive.
  - $\bar{V}$ is the long-run mean; a is the speed of mean reversion.
  - b is called the volatility of volatility.
  - If we set $a = b = 0$, we obtain the Black-Scholes model as a special case (with $\sigma = \sqrt{\bar{V}}$).

The Heston model and the smile

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- In practice, the standard procedure is to choose the parameters to achieve the best possible fit on the observed volatility surface, and then to use the calibrated model to price exotic options.
  - The important parameters are the correlation ρ and the volatility of volatility b.
- When $\rho = 0$, stochastic volatility ($b \neq 0$) increases the kurtosis of the risk-neutral distribution symmetrically (heavy tails). The higher the volatility of volatility, the more pronounced the smile.
- Negative correlation between the price of the underlying and volatility ($\rho < 0$) generates negative skewness, which translates into a downward-sloping volatility skew.

Beyond stochastic volatility models

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Estimating σ
Implied volatility
Implied risk-neutral distribution
▷ Volatility models
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- One well-known shortcoming of stochastic volatility models is that they cannot generate a large skew for short-maturity options.
- In order to account for option prices at very short maturities, people have developed models allowing for jumps in the underlying or even in volatility (so-called jump-diffusion models).
- There isn't a uniformly good model. Which model is right depends on the market under consideration.
- In fixed income markets, local volatility may be a reasonable model, since the volatility of interest rates is highly positively correlated with the level of interest rates.
- In FX markets, volatility is highly stochastic which means that (non-local) stochastic volatility models may give the best fit.
- In equity index markets, jump-diffusion models may be appropriate, especially in the short term.

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

Appendix A

Outline

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- Appendix A contains proofs of some results
- This material will not be examined

$\mathbb{E}[(X-K)^+]$ for X lognormal

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- To derive the Black-Scholes formula, we used the following result: Suppose X is lognormally distributed, and $\text{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 : \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \quad d_2 := d_1 - \sigma\sqrt{T-t}$$

- To simplify notation, let $\mathbb{E}(\log X) = \alpha$, and $\text{Var}(\log X) = \beta^2$ (i.e. $\beta := \sigma\sqrt{T-t}$)

Proof

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- We can write $\log X = \alpha + \beta z$, where z is standard normal. Thus $X = e^{\alpha+\beta z}$ and $\mathbb{E}(X) = e^{\alpha+\beta^2/2}$. Moreover, $X \geq K$ is equivalent to $\log X \geq \log K$,or

$$\begin{aligned} z &\geq \frac{\log K - \alpha}{\beta} \\ &= \frac{\log K - \log \mathbb{E}(X) + \beta^2/2}{\beta} \\ &= -\frac{\log[\mathbb{E}(X)/K] - \beta^2/2}{\beta} \\ &= -d_2. \end{aligned}$$

- Recall the standard normal density: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$
- Let Φ be the standard normal cdf.

Proof(cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

Thus we have:

$$\begin{aligned}
\mathbb{E}[(X-K)^+] &= \int_{-d_2}^{\infty} (e^{\alpha+\beta z} - K)\phi(z)dz \\
&= \int_{-d_2}^{\infty} e^{\alpha+\beta z}\phi(z)dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \int_{-d_2}^{\infty} \mathbb{E}(X)e^{-\frac{1}{2}\beta^2+\beta z}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2-\beta}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_1}^{\infty}\phi(z-\beta)d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2).
\end{aligned}$$

Breeden-Litzenberger

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q The price of a call option at $t = 0$ as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x - K)\varphi(x)dx$$

- Result [Breeden-Litzenberger (1978)]:

$$\varphi(K) = e^{rT}C''(K).$$

Proof of Breeden-Litzenberger result

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

Proof: We can write

$$e^{rT}C(K) = \int_K^\infty x\varphi(x)dx - K\int_K^\infty \varphi(x)dx.$$

Differentiating:

$$\begin{aligned} e^{rT}C'(K) &= \frac{d}{dK}\int_K^\infty x\varphi(x)dx - K\frac{d}{dK}\int_K^\infty \varphi(x)dx - \int_K^\infty \varphi \\ &= -K\varphi(K) - K[-\varphi(K)] - \int_K^\infty \varphi(x)dx \\ &= -\int_K^\infty \varphi(x)dx. \end{aligned}$$

Differentiating again:

$$e^{rT}C''(K) = -[-\varphi(K)] = \varphi(K).$$

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
▷ Appendix B

USEFUL FACTS

USEFUL FACTS

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
▷ Appendix B

- Put-call parity (non-dividend paying stock):

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

- Lognormal property of GBM:
  Suppose $dS_t = \mu S_t dt + \sigma S_t dW_t$. Then the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \sigma^2/2\right)(T-t), \sigma^2(T-t)\right]$$

- Black-Scholes formula:
  The price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2)$$

  where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

© S.Xiao (2025)

USEFUL FACTS

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
▷ Appendix B

- Black-Scholes PDE:
  Consider a derivative on a stock with payoff $g(S_T)$ at time T, whose price at time $t \leq T$ is given by $F(S,t)$. The Black-Scholes PDE is

$$F_t(S,t) + rSF_S(S,t) + \frac{1}{2}\sigma^2 S^2 F_{SS}(S,t) - rF(S,t) = 0$$

- Standard normal density:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Expectation of the exponential of a normal random variable:
  If X normally distributed, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\operatorname{Var}(X)\right]$$

© S.Xiao (2025)